Riad Mahmud Chowdhury

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CAREER OBJECTIVE

My goal is to grow as a researcher in Data Science, Machine Learning (ML), and Cybersecurity, working across the full data lifecycle: data cleaning, exploratory analysis, statistical testing, predictive modeling, and online learning. Building on my research in game-theoretic moving-target defense and cyber-physical network security, and on hands-on data projects at my own data science lab, I aim to develop data-driven methods for detecting attacks and anomalies, and robust, scalable, and energy-efficient defense algorithms.

RESEARCH INTERESTS

Machine Learning (ML), Data Science, Cybersecurity (Data-Driven Security & Anomaly Detection, Cyber-Physical Systems Security, Moving-Target Defense, Online Convex Optimization & Game Theory, Energy-Efficient ML), **Data Mining & Predictive Analytics** (Statistical Modeling, Regression, Hypothesis Testing, Data Visualization)

EDUCATION

Nantong University, School of Artificial Intelligence and Computer Science — **Nantong, Jiangsu, China**
Bachelor of Science Degree in Computer Science and Technology — 2023.09.10 – Present (Exp. 2027)
CGPA: **4.00** (Average Mark: **90.08 / 100**) (Completed 6th Semester / 144 Credits)

Chattogram Bandar College — **Chattogram, Bangladesh**
Higher Secondary Certificate (HSC) in Science — 2018 – 2020
Result: GPA: **4.87 / 5.00**

Mandarin Language Proficiency (HSK Level 4) — **Nantong University, China**
Result: **220 / 300** — 2025.12

SKILLS SUMMARY

- **Programming:** Python, R, JavaScript, C#, Java, SQL, HTML/CSS
- **Data Science & AI/ML:** Machine Learning, Linear Regression & Predictive Modeling, Online Convex Optimization (Entropic Mirror Descent), Parametric Linear Programming (LP), Game-Theoretic Modeling, Statistical Hypothesis Testing (χ^2 Detector), Exploratory Data Analysis (EDA), Data Cleaning, Time-Series Analysis (Resampling, Rolling Averages), Linear Algebra & Image Processing with NumPy
- **Libraries:** Pandas, NumPy, Scikit-learn, SciPy, Matplotlib, Seaborn, Plotly Express (Choropleth, Sunburst, Interactive Charts), Tidyverse, Ggplot2
- **Tools & Platforms:** pandapower, LP Solvers, Visual Studio Code, Google Colab, Jupyter Notebook, Git, GitHub (Organization Management), LaTeX, RStudio, Zotero, Microsoft Office
- **Languages:** Bengali (Native), English (Advanced / Fluent), Mandarin (Intermediate)

CONFERENCE PUBLICATIONS

- **International Conferences:**
  - **[IEEE CSDE'26]:** Md Mostofa Nurannabi Shakil, **Riad Mahmud Chowdhury**, Tanvir Hossain, Md. Alamgir Hossain, Md. Samiul Islam, and Md. Shafiqul Islam, "Provable Moving-Target Defense for Cyber-Physical Networks: From the Security–Performance Frontier to No-Regret Defense," in Proceedings of the 12th IEEE Asia-Pacific Conference on Computer Science and Data Engineering (IEEE CSDE 2026), Cox's Bazar, Bangladesh, November 1–3, 2026. [Accepted].

EXPERIENCE

Founder & Owner — DataPulse Data Science Lab — 2026 – Present
Lab GitHub: github.com/datapulse-lab-ds

- Founded and lead a collaborative data science lab focused on exploratory data analysis, applied statistics and hypothesis testing, interactive visualization, and predictive modeling.
- Manage the lab's GitHub organization, members, and repositories, and set the workflow for data cleaning, analysis notebooks, and project documentation.
- Led the lab's Movie Budget & Revenue Prediction project (see Projects).

Research Contributor & Co-Author (Cyber-Physical Systems & Data Analysis) — IEEE CSDE 2026

- **Experimental Modeling & Simulation:** Instantiated and validated a game-theoretic Moving-Target Defense (MTD) framework on **IEEE 14-bus (34 measurements)** and **IEEE 118-bus (304 measurements)** power grids using pandapower across **12 D-FACTS reactance configurations** (±60% range).
- **Security & Optimization Analysis:** Evaluated False-Data-Injection (FDI) stealth against DC Weighted-Least-Squares (WLS) state estimation and χ^2 residual detectors (5% false-alarm rate), empirically estimating confusion kernels (K) and validating parametric Linear Programming (LP) security–cost frontiers.
- **Algorithmic Evaluation & Scalability Impact:** Benchmarked entropic online mirror descent achieving sublinear $O(\sqrt{(T \log N)})$ regret over 6,000 epochs and demonstrated favorable grid scalability by reducing off-diagonal attack stealth from **0.14 to 0.04** and lowering the matrix-game security floor val(K) from **0.21 to 0.13**.

PROJECTS & SOFTWARE DEVELOPMENT

- **[Movie-Budget-Data-Science-project] Movie Budget & Revenue Prediction** (DataPulse Lab) 2026
  Cleaned dirty movie budget and revenue records (type conversion, missing values) and studied how film budgets changed over time. Trained a **Linear Regression** model (Scikit-learn, Seaborn) to predict worldwide box-office revenue from production budget.
- **[nobel-laureates-insights] Nobel Prize Historical Data Analysis** 2026
  Analyzed Nobel Prize laureates from 1901 to 2020 (Pandas, Seaborn, Plotly). Explored prize sharing over time, gender and country distribution (USA leads with **281** prizes), top research institutions and cities, and laureate age, using rolling averages, a choropleth world map, and a sunburst chart.
- **[android-app-store-analysis] Google Play Store App Analytics** 2026
  Cleaned and analyzed thousands of Android apps (Pandas, Plotly). Compared free vs. paid apps, found high-competition and high-opportunity categories, detected price outliers and junk apps, and estimated revenue for top-grossing apps.
- **[google-trends-data-analysis] Google Search Trends Analysis** 2026
  Analyzed Google search-trend time series with Pandas and Matplotlib, using date conversion, resampling, and rolling averages to compare search interest with real-world data on multi-axis charts.
- **[numpy-multidimensional-arrays] Computation with NumPy & N-Dimensional Arrays** 2026
  Worked with NumPy ndarrays, slicing, linear algebra (matrix multiplication), and broadcasting. Processed images as 3D arrays: grayscale conversion with luminance weights and image inversion.
- **[StackOverflow-Language-Trends] Developer Adoption Analytics** 2026
  Built a time-series analysis of SQL-extracted StackExchange data (**1,991 monthly records**, 14 languages, 2008–2020) with pivot tables and 12-month rolling means. JavaScript is the all-time leader, while Python became the #1 monthly language by 2020.
- **[college-salaries-analysis-pandas] Post-Graduation Compensation Analysis** 2026
  Analyzed PayScale salary data for **50 majors** (STEM, Business, HASS) with Pandas. Found the top starting and mid-career majors, the lowest-risk and highest-upside careers, and showed that STEM leads in mid-career median pay (**$90,812.50**).
- **[lego-data-analysis] Multi-Decade LEGO Dataset Insights** 2026
  Merged relational tables of **15,710 LEGO sets** (1949–2021) and 596 themes in Pandas and Matplotlib. Found an **18× rise** in average set size (12.36 parts in 1954 to 221.84 in 2017) and Star Wars as the top theme (753 sets).
- **[Superstore-Sales-Profitability-Analysis] Retail Sales Analytics in R** 2026
  Built a modular R workflow (Tidyverse, Ggplot2) to clean retail records and report profit by sub-category, region, shipping mode, and customer segment, showing how heavy discounts reduce net profit.
- **[Banking-Web-C-] Online Banking Web System** Dec 2025
  Built a full-stack banking web application in C# with object-oriented transaction logic, customer account management, database connectivity, and input validation.

CERTIFICATIONS & PROFESSIONAL TRAINING

- **Data, Data, Everywhere — Google Data Analytics Professional Certificate** Coursera | 2026
  Credential: Certificate Verification Link [Verified / Completed]
- **100 Days of Code: The Complete Python Pro Bootcamp** Udemy | 2026
  Credential: Certificate Verification Link [Verified / Completed]
- **Research Course: Beginning to Advance (33rd Batch)** Research Help Bangladesh | 2026
  Credential: Certificate Verification Link [Verified / Completed]

ACADEMIC HONORS & EXTRA-CURRICULARS

- **Perfect Academic Standing (CGPA 4.00):** Ranked in top tier throughout all completed semesters at Nantong University, School of Artificial Intelligence and Computer Science (2023 – Present).
- **Academic Mentorship:** Assisted peer student groups in scientific Python environments, structured notebook practices, and data visualization setups.

ACADEMIC REFERENCES

Xiaolin Ju
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